LLM & Prompt Engineering

Unit-1

Foundations of Prompt Engineering

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What is Generative AI?

- Generative AI refers to a class of artificial intelligence systems designed to generate new content. This includes text, images, music, code, voice, and even videos, based on patterns learned from existing data.

- Unlike traditional AI that focuses on prediction or classification, generative AI creates data rather than just analyzing it.

Example 1: Text Generation

A generative AI like ChatGPT can generate coherent and human-like paragraphs of text:

- Prompt:
  - “Write a poem about the ocean.”
- Output:

"The ocean whispers tales untold,
In shades of blue and gleaming gold..."

This is generative because the AI is not selecting from pre-written answers but creating new text based on training data.

Example 2: Image Generation

Using tools like DALL·E, users can input:

- Prompt:
  - "A futuristic city in the clouds at sunset."
- Output:

A never-before-seen image is generated, visually representing the scene.

Example 3: Code Generation

Tools like GitHub Copilot or CodeWhisperer use generative AI to write code:

Prompt:

- "Create a Python function to check if a number is prime."

- Output:

```
def is_prime(n):
    if n <= 1:
        return False
    for i in range(2, int(n**0.5)+1):
        if n % i == 0:
            return False
    return True
```

What are LLMs?

Large Language Models (LLMs) are a subset of generative AI trained on **massive datasets of text** using **deep learning**, especially **transformer architectures** like GPT (Generative Pre-trained Transformer).

LLMs like GPT-4, BERT, LLaMA, and Claude are capable of:

- Understanding natural language
- Answering questions
- Summarizing long documents
- Translating languages
- Writing stories, poems, essays, and code

https://www.pluralsight.com/resources/blog/ai-and-data/what-are-transformers-generative-ai

Example 4: LLM in Action

- Prompt:

"Explain black holes to a 10-year-old."

- LLM Output:

“A black hole is like a big space vacuum. It's a place in space where gravity is so strong that nothing, not even light, can escape from it!”

This showcases language understanding, simplification, and generation.

Relationship Between Generative AI and LLMs

Generative AI	***LLMs***
Broader category (includes images, audio, video, etc.)	Specializes in text and language
Includes models like DALL·E (images), MusicLM (music), etc.	Includes GPT, BERT, Claude, etc.
Use case: generating artwork	Use case: writing essays, chatbots

Applications of Generative AI and LLMs

- **Education – Personalized tutoring, automated content creation**
- **Healthcare – Clinical documentation, medical chatbots**
- **Software Development – Code generation, bug fixing**
- **Media – Script writing, game dialogue creation**
- **Business – Report generation, customer service automation**

Challenges and Ethical Concerns

- **Bias and fairness: LLMs can reflect societal biases**
- **Misinformation: May generate false or misleading content**
- **Plagiarism & originality: Risk of copying training data**
- **Job displacement: Automation of creative tasks**

What is Prompt Engineering?

- Prompt Engineering is the art and science of crafting effective inputs (called "prompts") to get accurate, useful, or creative responses from AI models like ChatGPT, Google Gemini, Claude, or any Large Language Model (LLM).

- Think of prompt engineering like giving instructions to a very smart assistant—you need to be clear, precise, and intentional to get the best outcome.

Why is Prompt Engineering Important?

- Large Language Models don’t "understand" in the human sense. They generate answers based on patterns learned during training. So:
- A well-crafted prompt → gives high-quality, relevant output.
- A vague or unclear prompt → leads to poor or random responses.

Example: Simple vs. Smart Prompt:

- ❌ Poor Prompt:
  - Write something about India.
- ✅ Good Prompt:
  - Write a 150-word paragraph about the cultural diversity of India, focusing on languages, festivals, and traditions, suitable for a school textbook.

  - ✅ Result: You get focused, relevant, and structured output.

Structure of a Prompt:

- Understanding the structure of a good prompt is essential in Prompt Engineering.
- A well-structured prompt helps Large Language Models (LLMs) like ChatGPT understand your expectations and generate high-quality responses.
- A good prompt typically has the following components:

Component	Description
1. Role	Who should the AI act as? Optional but useful for setting tone and expertise.Example: "Act as a career counselor."
2. Task	What should the AI do? Clearly define the expected action or objective.Example: "Suggest five career paths for a computer science graduate."
3. Context	Background information or data needed to understand the task.Example: "The student prefers remote work and has skills in Python and ML."
4. Format/Output	What format should the output take?(e.g., bullet list, paragraph, code block, table)Example: "Provide a bullet list with a one-line explanation for each career."
5. Constraints	Any restrictions like word limit, language complexity, tone, or style.Example: "Use simple language suitable for high school students. Keep each response under 100 words."
6. Examples	(Optional) Demonstrate the expected format/output (few-shot learning)._Example: Input: "Suggest careers for a biology student. Output: 1. Biotechnologist – works with living systems in R&D."

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1. Role **Helps guide the tone and domain.** ➢ **Example: “You are a Python tutor.”** **2. Task** **Be clear about what you want the AI to do.** ➢ **Example: “Explain list comprehension in simple terms.”** **3. Context** **Give background info or data if needed.** ➢ **Example: “Based on the below text…”**	**4. Format** **Specify the structure you expect.** ➢ **Example: “Provide your answer in 3 bullet points.”** **5. Constraints** **Limitations like tone, length, or audience.** ➢ **Example: “Answer in under 100 words.”** **6. Examples (Few-shot prompting)** **Use this when format/style is complex or specific.** ➢ **Example:** **a). Question: What is AI? → Answer: AI is the simulation of human intelligence in machines.** **b). Question: What is ML? → Answer:**

Structure of a Prompt:

You are a science teacher.

Explain the concept of gravity to a 12-year-old student using simple language and one example. Keep the explanation under 50 words.

✅ **This prompt clearly sets role, task, context, output format, and constraints.**

Prompt Engineering Techniques:

1. Zero-Shot Prompting:

Definition:

You provide the AI with a task without any example—just an instruction.

Example Prompt:

Write a haiku about a peaceful forest.

Output:

Whispering pine trees

A deer steps on fallen leaves

Silence hugs the trail

Use When:

You expect the model to know the task from its training and the task is simple or common.

Prompt Engineering Techniques:

2. One-Shot Prompting:

Definition:

You give the AI one example to guide the expected format or logic.

Example Prompt:

Translate English to Marathi:

English: How are you? → Marathi:

English: I am fine → Marathi:

Output:

मी ठीक आहे

Use When:

You want the model to follow a specific style, pattern, or language.

Prompt Engineering Techniques:

3. Few-shot Prompting

Definition:

Few-shot prompting is a technique where the prompt includes a few examples of the task along with the actual input. This helps the model learn the pattern or format to follow.

Example Prompt:

Q: What is the capital of France?

A: Paris

Q: What is the capital of Germany?

A: Berlin

Q: What is the capital of Japan?

A:

Output:

Tokyo

Use When:

Use when the model needs to learn a pattern from examples, especially when fine-tuning is not feasible and zero-shot prompting gives inaccurate results.

Prompt Engineering Techniques:

4. Chain-of-Thought Prompting

Definition:

You instruct the AI to show step-by-step reasoning before giving the final answer.

Example Prompt:

A train travels 120 km in 3 hours. What is its speed? Show your steps.

Output:

Step 1: Distance = 120 km

Step 2: Time = 3 hours

Step 3: Speed = Distance ÷ Time = 120 ÷ 3 = 40 km/h

Answer: 40 km/h

Use When:

You want better accuracy in reasoning tasks (math, logic, multi-step problems).

Prompt Engineering Techniques:

5. Instruction-Based Prompting Definition:

Definition:

You write a clear instruction telling the AI exactly what to do.

Example Prompt:

Summarize this paragraph in 2 bullet points.

Use When:

The task is straightforward, but the format or boundaries are essential (length, tone, structure).

Prompt Engineering Techniques:

6. Role-Based Prompting

Definition:

You assign the AI a role to influence its tone, knowledge, and depth.

Example Prompt:

You are a history teacher. Explain the importance of the French Revolution to a 10th-grade student.

Output:

The French Revolution was important because it ended the rule of kings in France and introduced the idea that people should have rights and a say in government.

Use When:

You want domain-specific or audience-specific answers (doctor, lawyer, child, expert, etc.).

Prompt Engineering Techniques:

7. Contextual Prompting

Definition:

You provide extra information or background context before the task.

Example Prompt:

Based on the following text, write a headline:\n\n"Apple released its new iPhone with a foldable screen and improved battery life."\n\nPrompt: Write a headline.

Output:

Apple Unveils Foldable iPhone with Longer Battery Life

Use When:

You want output based on a given text or document.

Prompt Engineering Techniques:

8. Multi-Turn Prompting (Conversational)

Definition:

AI is guided across multiple questions and responses, like a dialogue or iterative task.

Example Prompt:

User: Explain quantum computing simply.

AI: (Provides a simple explanation)

User: Can you give an analogy?

AI: (Provides analogy)

User: Now give a real-world use case.

Use When:

You're interacting in stages and refining the response interactively.

Technique	Description	Best For
Zero-Shot Prompting	Direct instruction, no examples	Simple, known tasks
One-Shot Prompting	One example provided	Basic format training
Few-Shot Prompting	Multiple examples provided	Repetitive patterns or format-specific output
Chain-of-Thought	Step-by-step reasoning	Math, logic, and complex tasks
Instruction-Based	Clear task instructions	Summarization, formatting, rewriting
Role-Based Prompting	Assign AI a role	Domain-specific tone and answers
Contextual Prompting	Background or data included	Data-driven or content-specific tasks
Multi-Turn Prompting	Step-by-step interaction	Interactive sessions and refinement